Layer 9: Surface Enzyme Kinetics

Cleavage Kinetics of Surface-Immobilised SGKIPRTLTA by MMP-9

Eton iGEM 2026, Dry Lab | Layer 9 of 17-Layer Biosensor Pipeline | `surface_kinetics.py`

I. Physical Picture

The biosensor probe SGKIPRTLTA is surface-immobilised on the gold electrode via a Cys-thiol anchor and carries a methylene blue (MB) redox label at the C-terminal lysine. At surface concentration Γ_0 (mol cm^{-2}), the intact probe keeps MB tethered close to the electrode; MMP-9 cleavage releases the MB-bearing fragment into solution, decreasing the SWV peak current. The fractional cleavage $F_k(t)$ is therefore the central output of this layer and the direct precursor of the ratiometric signal $\Delta R/R_0$ (Layer 13).

The governing kinetics are heterogeneous: the enzyme (E, MMP-9) diffuses freely in solution while the substrate (S, probe) is fixed at the surface. Classical solution-phase Michaelis–Menten equations, derived under the condition $[S] \gg [E]$, do not apply in this configuration because the total molar quantity of surface-immobilised peptide ($\Gamma_0 \cdot A \sim 10$ fmol) can be comparable to or less than the bulk enzyme content. Anne and Demaille [1] derived exact kinetic equations for this heterogeneous surface mechanism without invoking the steady-state approximation.

The enzyme input $[E]_t = [\text{MMP-9}]_{\text{free}}$ is taken directly from the Layer 0 wound biology ODE output: 30 nM (chronic wound, 30-min equilibration) and 6 nM (healing wound). TIMP-1 inhibition is fully accounted for in these values; no further correction is needed here.

II. Three-State Surface ODE System

Reaction mechanism

The surface enzyme reaction follows the Michaelis–Menten mechanism with the substrate anchored to the surface [1]:

$$S + E \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_{\text{cat}}} P + E \tag{II.1}$$

where S, ES, P are surface species with concentrations Γ_S, Γ_{ES}, Γ_P (mol cm^{-2}); k_1 (M^{-1}s^{-1}) is the second-order binding rate constant; k_{-1} (s^{-1}) is the dissociation rate constant; and k_{cat} (s^{-1}) is the catalytic rate constant. The enzyme concentration at the electrode surface $[E]_t$ (M) is treated as constant throughout (justified in Section VI).

Governing ODEs

The time evolution of each surface species is [1] (Eq. 1):

$$\frac{d\Gamma_S}{dt} = k_{-1}\,\Gamma_{ES} - k_1[E]_t\,\Gamma_S, \qquad \frac{d\Gamma_{ES}}{dt} = k_1[E]_t\,\Gamma_S - (k_{-1} + k_{\text{cat}})\,\Gamma_{ES}, \qquad \frac{d\Gamma_P}{dt} = k_{\text{cat}}\,\Gamma_{ES} \tag{II.2}$$

with surface conservation and initial conditions:

$$\Gamma_S + \Gamma_{ES} + \Gamma_P = \Gamma_0, \qquad \Gamma_S(0) = \Gamma_0, \quad \Gamma_{ES}(0) = \Gamma_P(0) = 0 \tag{II.3}$$

Introducing the unconverted substrate $\Gamma^* = \Gamma_S + \Gamma_{ES}$ and defining $k_1^* = k_1[E]_t$, the system reduces to the two-variable form used by Anne and Demaille [1] for the Laplace analysis:

$$\frac{d\Gamma_{ES}}{dt} = k_1^*\,\Gamma^* - (k_1^* + k_{-1} + k_{\text{cat}})\,\Gamma_{ES}, \qquad \frac{d\Gamma^*}{dt} = -k_{\text{cat}}\,\Gamma_{ES} \tag{II.4}$$

III. Analytic Solutions

Steady-state approximation

Setting $d\Gamma_{ES}/dt = 0$ in Eq. (II.4) and substituting the conservation constraint yields the steady-state progress curve [1] (Eq. 3):

$$\theta_P^{\text{ss}} = \frac{\Gamma_P}{\Gamma_0} = 1 - \exp(-k_{\text{eff}}\,t), \qquad k_{\text{eff}} = \frac{k_{\text{cat}}}{1 + K_M/[E]_t}, \quad K_M = \frac{k_{-1} + k_{\text{cat}}}{k_1} \tag{III.1}$$

This is the Langmuirian (LG) progress curve. The denominator $(1 + K_M/[E]_t)$ shows that k_{eff} can only reach its maximum k_{cat} when $[E]_t \gg K_M$ (enzyme-saturated regime). At clinical concentrations $[E]_t \leq 30$ nM $\ll K_M \approx 1.28$ mM [2] (Section V), the denominator $\approx K_M/[E]_t$ and the expression simplifies to: $k_{\text{eff}} \approx (k_{\text{cat}}/K_M)[E]_t \equiv k_{\text{eff}}^{\text{surf}}[E]_t$. In this limit $[E]_t$ appears only in the numerator and the rate is first-order in enzyme: the enzyme is in excess and does not appear in any denominator in the operative equations.

Exact biexponential solution

The reduced system Eq. (II.4) is linear and solved exactly via Laplace transform [1]. Define the composite parameters:

$$k_1^* = k_1[E]_t, \qquad \sigma = k_1^* + k_{-1} + k_{\text{cat}}, \qquad \Delta = \sigma^2 - 4k_1^* k_{\text{cat}} > 0 \tag{III.2}$$

The exact solutions are [1] (Eqs. 4–5):

$$\Gamma_{ES}(t) = \frac{k_1^*\,\Gamma_0}{\sqrt{\Delta}}\left[\exp\left(-\frac{\sigma-\sqrt{\Delta}}{2}\,t\right) - \exp\left(-\frac{\sigma+\sqrt{\Delta}}{2}\,t\right)\right] \tag{III.3}$$

$$\Gamma^*(t) = \frac{\Gamma_0}{2\sqrt{\Delta}}\left[(\sigma+\sqrt{\Delta})\exp\left(-\frac{\sigma-\sqrt{\Delta}}{2}\,t\right) - (\sigma-\sqrt{\Delta})\exp\left(-\frac{\sigma+\sqrt{\Delta}}{2}\,t\right)\right] \tag{III.4}$$

The fractional product coverage (fractional cleavage) is then:

$$F_k(t) = \theta_P(t) = 1 - \frac{\Gamma^*(t)}{\Gamma_0} \tag{III.5}$$

λ parameter and dimensionless universal form

Anne and Demaille [1] introduce the dimensionless kinetic zone parameter:

$$\lambda = \frac{\sigma}{\sqrt{\Delta}} = \frac{1}{\sqrt{1 - 4k_1^* k_{\text{cat}}/\sigma^2}} = \frac{1}{\sqrt{1 - 4k_1 k_{\text{cat}}[E]_t\,/\,(k_1^* + k_{-1} + k_{\text{cat}})^2}} \geq 1 \tag{III.6}$$

and the dimensionless time $\tau = \sigma(1 - 1/\lambda)\,t$. In these variables the exact progress curve Eq. (III.5) becomes [1] (Eq. 7):

$$\theta_P(\tau,\lambda) = 1 - \frac{1}{2}\left\{(1+\lambda)\exp\left[-\frac{\tau}{2}\right] + (1-\lambda)\exp\left[-\frac{1}{2}\frac{\lambda+1}{\lambda-1}\,\tau\right]\right\} \tag{III.7}$$

This single equation captures the entire range of possible kinetic behaviours through the value of λ alone [1]. Two limiting cases are:

- **LG limit** ($\lambda \to 1$): Eq. (III.7) reduces to $\theta_P = 1 - \exp(-\tau/2)$, a simple monoexponential. The steady-state approximation is exact. Diagnostic: $t_{1/2}/t_{1/4} \to 2.41$ [1].
- **FS limit** ($\lambda \gg 1$): Eq. (III.7) reduces to [1] (Eq. 8):

$$\theta_P = 1 - \left(1 + \frac{\tau}{2}\right)\exp\left(-\frac{\tau}{2}\right) \tag{III.8}$$

  The progress curve is S-shaped (sigmoidal foot at short times). The steady-state approximation overestimates θ_P and fails here. Diagnostic: $t_{1/2}/t_{1/4} \to 1.75$ [1].

The zone diagram [1] (Fig. 4) maps these regimes in $\log(k_{-1}/k_{\text{cat}}) - \log([E]_t/K_M)$ space. The LG zone applies when $k_{-1}/k_{\text{cat}} \gg \Phi$, where $\Phi = 4[E]_t/K_M\,/(1 + [E]_t/K_M)^2$, or unconditionally when $[E]_t/K_M \leq 0.2$ or $[E]_t/K_M \geq 5$.

IV. Surface Probe Saturation: The Integrated MM Equation

A second, independent saturation condition

The A&D analysis in Sections II–III characterises the kinetic zone along the *enzyme* axis: does the enzyme–probe complex form and dissociate fast enough that the steady-state approximation holds? The conclusion $\lambda \approx 1$ (Section V) is a statement about enzyme-side kinetics and holds independently of the probe surface coverage.

A second condition must be checked separately: whether the probe density Γ_0 is large or small relative to the *surface* Michaelis constant $K_{m,\text{surf}}^{\Gamma}$ (units: mol cm^{-2}). The A&D Eqs. 4–7 assume the probe is always in the unsaturated (dilute) regime, i.e. effectively $K_{m,\text{surf}}^{\Gamma} \to \infty$. At the surface coverages used in our device this assumption requires verification.

The heterogeneous surface MM ODE with correct units

The surface Michaelis–Menten equation governing probe depletion, with substrate and product both expressed as surface concentrations (mol cm^{-2}) and enzyme as a volume concentration (mol L^{-1}), is (Squires et al. 2008 [4], Eq. 1):

$$\frac{d\Gamma_S}{dt} = -\frac{k_{\text{cat}}\,[E]_t\,\Gamma_S}{K_{m,\text{surf}}^{\Gamma} + \Gamma_S} \tag{IV.1}$$

In the A&D notation, $K^{\Gamma}_{m,\mathrm{surf}}$ is the probe density at which the per-probe turnover rate is half its maximum value. Its units are mol cm^{-2}, not mol L^{-1}. The enzyme concentration $[E]_t$ appears only in the numerator as a rate multiplier; the denominator contains surface concentrations only ($K^{\Gamma}_{m,\mathrm{surf}}$ and Γ_S). This reflects the physical situation: MMP-9 is present in bulk excess in solution and is *not* the limiting species. The denominator $K^{\Gamma}_{m,\mathrm{surf}}+\Gamma_S$ captures saturation of enzyme–probe encounters at high probe density, not saturation of the enzyme active sites (which would require $[E]_t$ in the denominator).

Conversion from solution K_m to surface units

The solution-phase K_m^{sol} (mol L^{-1}, confirmed for SGKIPRTLTA from Kridel 2001 [2]) is related to the surface saturation constant by the effective thickness of the accessible probe layer δ_{eff} (Layer 8):

$$K^{\Gamma}_{m,\mathrm{surf}} = K_m^{\mathrm{sol}}\,[\mathrm{mol\ L^{-1}}] \times 10^{-3}\,[\mathrm{L\ cm^{-3}}] \times \delta_{\mathrm{eff}}\,[\mathrm{cm}] \tag{IV.2}$$

For $K_m^{\mathrm{sol}} = 1.28$ mM and $\delta_{\mathrm{eff}} = 5$ nm $= 5\times10^{-7}$ cm (Layer 8 probe extent, unconfirmed):

$$K^{\Gamma}_{m,\mathrm{surf}} = 1.28\times10^{-3}\times10^{-3}\times5\times10^{-7} = 6.4\times10^{-13}\ \mathrm{mol\ cm^{-2}} \tag{IV.3}$$

The probe-saturation ratio ρ

Define the dimensionless probe-saturation parameter:

$$\rho = \frac{\Gamma_0}{K^{\Gamma}_{m,\mathrm{surf}}} \tag{IV.4}$$

ρ controls the shape of the progress curve independently of λ (which controls the enzyme-saturation regime). Two limits:

- $\rho \ll 1$: probe is always in the unsaturated regime $\Gamma_S \ll K^{\Gamma}_{m,\mathrm{surf}}$; Eq. (IV.1) reduces to first-order in Γ_S; the monoexponential Eq. (V.3) is recovered with $k_{\mathrm{eff}}^{\mathrm{surf}} = k_{\mathrm{cat}}/K_m^{\mathrm{sol}}$.
- $\rho \gg 1$: the probe is initially in the zeroth-order (saturated) regime; the cleavage rate is independent of Γ_S and proportional only to $[E]_t$: $d\Gamma_S/dt = -k_{\mathrm{cat}}[E]_t \to F_k = k_{\mathrm{cat}}[E]_t\,T/\Gamma_0$. The progress curve is *linear* in time, with a Langmuir-like saturation shape at late times as Γ_S falls through $K^{\Gamma}_{m,\mathrm{surf}}$.
- $\rho \sim 1$: intermediate (physically realistic) case: the progress curve is *between* the monoexponential and the Langmuir-like linear regime, as treated by the integrated equation below.

At the unconfirmed reference values ($\Gamma_0 = 2\times10^{-12}$ mol cm^{-2} from Pheeney 2012 [5] midpoint; no steric correction): $\rho_0 = 2\times10^{-12}/6.4\times10^{-13} \approx 3.1$. With the Layer 8 steric correction (which increases $K^{\Gamma}_{m,\mathrm{surf}}$ by f_{steric}^{-1}; unconfirmed factor $\approx$ 2–10×):

f_{steric} (Layer 8)	$K^{\Gamma}_{m,\mathrm{surf}}$ (mol cm^{-2})	ρ	Dominant regime
0.1 (high crowding)	6.4×10^{-12}	0.31	near first-order
0.3	2.1×10^{-12}	0.95	intermediate
0.5	1.3×10^{-12}	1.5	intermediate
1.0 (no correction)	6.4×10^{-13}	3.1	intermediate–zeroth

Table 1: Probe-saturation ratio ρ across the plausible steric correction range from Layer 8. $\rho = 1$ is the critical point at which the progress curve deviates most from the monoexponential. The system is unlikely to be in the pure monoexponential limit ($\rho \ll 1$) at clinical probe densities unless crowding is severe ($f_{\mathrm{steric}} < 0.05$).

Implicit integrated Michaelis–Menten solution

Integrating Eq. (IV.1) from $\Gamma_S = \Gamma_0$ to $\Gamma_S = \Gamma_0(1 - F_k)$, with $[E]_t$ constant (justified by Da $\ll 1$, Section VI):

$$\boxed{k_{\mathrm{cat}}\,[E]_t\,t = K^{\Gamma}_{m,\mathrm{surf}}\,\ln\frac{1}{1-F_k} + \Gamma_0\,F_k} \tag{IV.5}$$

This is the primary model for $F_k(T)$. It is an implicit equation in F_k and must be inverted numerically (bisection or Newton's method, implemented in `surface_kinetics.py`). In dimensionless form, with $\tilde{t} = k_{\mathrm{cat}}[E]_t\,t/K^{\Gamma}_{m,\mathrm{surf}}$ and $u = F_k$:

$$\tilde{t} = -\ln(1-u) + \rho\,u \tag{IV.6}$$

Recovery of the two limits:

$$\rho = 0: \quad \tilde{t} = -\ln(1-u) \;\Rightarrow\; F_k = 1 - \exp\!\left(-\frac{k_{\text{cat}}}{K_m^{\text{sol}}}[E]_t\, t\right) \quad \text{(monoexponential)} \tag{IV.7}$$

$$\rho \to \infty: \quad \tilde{t} \approx \rho\, u \;\Rightarrow\; F_k = \frac{k_{\text{cat}}[E]_t\, t}{\Gamma_0} \quad \text{(Langmuir-like, linear in } t\text{)} \tag{IV.8}$$

The two-parameter form $\{k_{\text{cat}}[E]_t,\, K_{m,\text{surf}}^{\Gamma}\}$ (equivalently $\{k_{\text{obs}},\, \rho\}$) is fit to experimental progress curves from a single incubation time-series. The monoexponential (one-parameter fit) is recovered as the $\rho \to 0$ special case.

Short-time approximation and discrimination ratio

At early times ($F_k \ll 1$), Taylor-expanding Eq. (IV.6): $\tilde{t} \approx (1+\rho)\, u + \ldots$, so:

$$F_k(T) \approx \frac{k_{\text{cat}}[E]_t\, T}{K_{m,\text{surf}}^{\Gamma} + \Gamma_0} = \frac{k_{\text{cat}}[E]_t\, T}{K_{m,\text{surf}}^{\Gamma}(1+\rho)} \quad \text{(short-}T\text{ linear regime)} \tag{IV.9}$$

The ratio of chronic to healing signal at short times is independent of both k_{cat} and $K_{m,\text{surf}}^{\Gamma}$ (and of ρ):

$$\frac{F_k^{\text{chronic}}}{F_k^{\text{healing}}} \approx \frac{[E]_{\text{chronic}}}{[E]_{\text{healing}}} = 5\times \tag{IV.10}$$

This result, previously derived from the monoexponential limit, holds for *all* ρ in the short-time regime. The 5× discrimination ratio is therefore robust to the value of ρ at early incubation times.

V. Kinetic Regime at Clinical MMP-9 Concentrations

$\lambda \approx 1$: the Langmuirian limit is guaranteed

At clinical concentrations $[E]_t \in [0.19,\, 2.17]$ nM and any physically realisable binding rate constant $k_1 \leq 10^8$ $\text{M}^{-1}\text{s}^{-1}$ (diffusion limit for a 92 kDa enzyme):

$$k_1^* = k_1[E]_t \leq 10^8 \times 2.17 \times 10^{-9} = 0.22\ \text{s}^{-1} \tag{V.1}$$

The solution-phase value $k_{\text{cat}}^{\text{sol}} = 86.4\ \text{s}^{-1}$ [2] is an upper bound for the surface k_{cat}. In the regime $k_1^* \ll k_{\text{cat}}$:

$$\Delta \approx (k_{-1}+k_{\text{cat}})^2 + 2k_1^*(k_{-1}-k_{\text{cat}}) \approx (k_{-1}+k_{\text{cat}})^2, \quad \lambda = \frac{k_1^* + k_{-1} + k_{\text{cat}}}{\sqrt{\Delta}} \approx 1 + O\!\left(\frac{k_1^*}{k_{\text{cat}}}\right) \tag{V.2}$$

Since $k_1^*/k_{\text{cat}}^{\text{sol}} \leq 2.5 \times 10^{-3}$, we have $\lambda - 1 < 3 \times 10^{-3}$ regardless of the unknown surface k_{-1}. The system is firmly in the LG zone.

Collapse to monoexponential

In the LG limit, the exact biexponential Eq. (III.7) degenerates to the Langmuirian monoexponential irrespective of the k_{-1}/k_{cat} ratio:

$$\boxed{F_k(t) = 1 - \exp\!\left(-k_{\text{eff}}^{\text{surf}}\,[E]_t\, t\right)} \tag{V.3}$$

where $k_{\text{eff}}^{\text{surf}} \equiv k_{\text{cat}}^{\text{surf}}/K_M^{\text{surf}}$ is the single measurable surface rate constant. This result is notable: the unknowns k_1, k_{-1}, and $k_{\text{cat}}^{\text{surf}}$ individually are inaccessible at clinical $[E]_t$, but the product $k_{\text{eff}}^{\text{surf}}[E]_t$ is directly fit from the progress curve. The complex accumulates at sub-ppm levels ($\Gamma_{ES}/\Gamma_0 < 10^{-3}$) and is kinetically invisible.

The solution upper bound is $k_{\text{eff}}^{\text{surf}} \leq k_{\text{cat}}^{\text{sol}}/K_m^{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ [2]. Surface immobilisation reduces this by 2–100× depending on steric accessibility (Layer 8); the surface value must be measured on the fabricated electrode.

VI. Dimensionless Analysis

Three dimensionless numbers characterise the device operating regime and validate the sequential layer architecture.

Kinetic zone parameter λ

From Section V: $\lambda \approx 1$ for all clinical $[E]_t$. The monoexponential Eq. (V.3) is the appropriate model. The zone parameter provides an experimental diagnostic: if the measured progress curve shows $t_{1/2}/t_{1/4}$ significantly below 2.41, the system has left the LG zone and the full exact solution Eq. (III.7) must be used.

Damköhler number: transport does not limit

The Damköhler number compares the surface reaction rate to the enzyme delivery rate by diffusion [4]:

$$\mathrm{Da} = \frac{k_{\mathrm{eff}}^{\mathrm{surf}}\,[E]_t\,h}{D_E} \tag{VI.1}$$

where $h \approx 100$ μm is the diffusion layer height (Layer 3) and $D_E \approx 4.7 \times 10^{-7}$ cm^{2}s^{-1} is the MMP-9 diffusion coefficient (Stokes–Einstein, Layer 3). At the solution upper bound with $[E]_t = 2.17$ nM:

$$\mathrm{Da}^{\max} = \frac{67{,}500 \times 2.17 \times 10^{-9} \times 10^{-2}}{4.7 \times 10^{-7}} \approx 3.1 \times 10^{-3} \ll 1 \tag{VI.2}$$

The electrode surface sees the bulk enzyme concentration to within $< 0.4\%$ at all times. For any plausible surface $k_{\mathrm{eff}}^{\mathrm{surf}} < 67{,}500$ M^{-1}s^{-1}, Da is smaller still. The Layer 3 mass-transport correction is negligible; $[E]_t$ in Eq. (V.3) is the bulk concentration from Layer 0 throughout.

The Damköhler criterion is consistent with the general analysis of Squires et al. [4]: for $\mathrm{Da} \ll 1$ the sensor operates in the reaction-limited regime where signal is proportional to the surface reaction rate, not the diffusion flux, and the response is linear in enzyme concentration at short times.

Enzyme conservation: $[E]_t$ is constant

MMP-9 is a true catalyst: it is released intact after each cleavage event (Eq. (II.1)). The bulk enzyme concentration $[E]_t$ therefore does not decrease as Γ_P increases. This is distinct from, for example, irreversible inhibitors. The constancy of $[E]_t$ is a consequence of the catalytic mechanism, not an approximation, and holds for all t regardless of F_k.

VII. Coupled PDE–ODE Numerical Treatment

Full coupled system

For completeness, and to verify the $\mathrm{Da} \ll 1$ approximation, the full coupled system – Layer 3 diffusion PDE and Layer 9 surface ODE – is solved simultaneously:

$$\frac{\partial [E]}{\partial t} = D_E\,\frac{\partial^2 [E]}{\partial z^2}, \quad 0 < z < h \tag{VII.1}$$

$$\frac{d\Gamma_{ES}}{dt} = k_1^*\Gamma^* - (k_1^* + k_{-1} + k_{\mathrm{cat}})\Gamma_{ES} \quad \text{with } k_1^* = k_1[E](z{=}0,t) \tag{VII.2}$$

with boundary conditions:

$$D_E\,\frac{\partial [E]}{\partial z}\bigg|_{z=0} = k_{\mathrm{cat}}\,\Gamma_{ES}(t) \quad \text{(flux = cleavage consumption)}, \qquad [E](h,t) = [E]_{\mathrm{bulk}} \tag{VII.3}$$

Solved numerically: Crank–Nicolson finite difference for the PDE, RK4 for the surface ODE, coupled at each timestep through the $z = 0$ boundary condition.

Verification of the decoupled limit

For $\mathrm{Da} \ll 1$, the PDE solution gives $[E](0,t) \approx [E]_{\mathrm{bulk}}$ to within $\mathrm{Da}/(1+\mathrm{Da}) < 0.4\%$. The surface ODE then has a constant-coefficient enzyme input and reduces exactly to Eq. (V.3). The coupled simulation (`surface_kinetics.py`) confirms this: the full PDE–ODE $F_k(t)$ is indistinguishable from the analytic monoexponential at all clinical $[E]_t$ values. The decoupled monoexponential is therefore used for all subsequent analysis.

VIII. $F_k(t)$ Output and Optimal Readout Time

Progress curves

From Eq. (V.3), the fractional cleavage at incubation time T:

$$F_k(T) = 1 - \exp\!\left(-k_{\mathrm{eff}}^{\mathrm{surf}}\,[E]_t\,T\right) \tag{VIII.1}$$

In the linear regime ($k_{\mathrm{eff}}^{\mathrm{surf}}[E]_t T \ll 1$):

$$F_k(T) \approx k_{\mathrm{eff}}^{\mathrm{surf}}\,[E]_t\,T \tag{VIII.2}$$

The ratio of chronic to healing fractional cleavage is:

$$\frac{F_k^{\mathrm{chronic}}}{F_k^{\mathrm{healing}}} \approx \frac{[E]_{\mathrm{chronic}}}{[E]_{\mathrm{healing}}} = \frac{30\ \mathrm{nM}}{6\ \mathrm{nM}} = 5\times \tag{VIII.3}$$

independent of $k_{\mathrm{eff}}^{\mathrm{surf}}$ and T. The $5\times$ discrimination ratio is therefore robust across the entire plausible range of surface rate constants.

Optimal readout time T^*

In the linear regime, $F_k \propto [E]_t T$, so the ratio F_k^c/F_k^h is always 5× regardless of T. The optimal incubation time is then as long as possible before signal saturation ($F_k \to 1$).

In the exponential (nonlinear) regime, the *absolute* signal difference $\Delta F_k = F_k^{\text{chronic}} - F_k^{\text{healing}}$ varies with T. Maximising ΔF_k gives the optimal readout time. Setting $d(\Delta F_k)/dT = 0$:

$$k_{\text{eff}}^{\text{surf}}\,[E]_c\,\exp\!\big(-k_{\text{eff}}^{\text{surf}}[E]_c T\big) = k_{\text{eff}}^{\text{surf}}\,[E]_h\,\exp\!\big(-k_{\text{eff}}^{\text{surf}}[E]_h T\big)$$
$$\Rightarrow \quad T^* = \frac{\ln([E]_c/[E]_h)}{k_{\text{eff}}^{\text{surf}}\,([E]_c - [E]_h)} \tag{VIII.4}$$

With $[E]_c = 30$ nM, $[E]_h = 6$ nM, and ratio $r = 5$:

$$T^* = \frac{\ln 5}{k_{\text{eff}}^{\text{surf}} \times 24\ \text{nM}} = \frac{1.609}{k_{\text{eff}}^{\text{surf}} \times 2.4 \times 10^{-8}\ \text{M}} \tag{VIII.5}$$

$k_{\text{eff}}^{\text{surf}}$ ($\text{M}^{-1}\text{s}^{-1}$)	T^* (s)	T^* (min)
2,000	33,500	558
10,000	6,700	112
30,000	2,230	37
67,500 (sol. UB)	990	16

Table 2: Optimal readout time T^* (Eq. (VIII.5)) as a function of the surface rate constant $k_{\text{eff}}^{\text{surf}}$. The 30–60 min clinical window corresponds to $k_{\text{eff}}^{\text{surf}} \approx 13{,}000$–$27{,}000\ \text{M}^{-1}\text{s}^{-1}$.

The target incubation window of 30–60 min corresponds to $k_{\text{eff}}^{\text{surf}} \approx 13{,}000$–$27{,}000\ \text{M}^{-1}\text{s}^{-1}$. If the measured $k_{\text{eff}}^{\text{surf}}$ falls significantly below this, the incubation time must be extended accordingly.

IX. Temperature Dependence

Arrhenius correction

The surface rate constant follows Arrhenius temperature dependence:

$$k_{\text{eff}}^{\text{surf}}(T_{\text{amb}}) = k_{\text{eff}}^{\text{surf}}(T_{\text{ref}})\,\exp\!\left[-\frac{E_a}{R}\left(\frac{1}{T_{\text{amb}}} - \frac{1}{T_{\text{ref}}}\right)\right] \tag{IX.1}$$

where E_a is the activation energy and $T_{\text{ref}} = 310$ K (37°C). The relative sensitivity to ambient temperature is:

$$\frac{1}{k_{\text{eff}}}\frac{dk_{\text{eff}}}{dT_{\text{amb}}} = \frac{E_a}{R\,T_{\text{amb}}^2} \tag{IX.2}$$

The activation energy E_a for MMP-9 surface cleavage of SGKIPRTLTA is not confirmed from any paper in `papers/layer_09/`. Using a typical metalloproteinase value $E_a \approx 45$ kJ mol^{-1} (unconfirmed placeholder):

$$\left.\frac{d\ln k_{\text{eff}}}{dT}\right|_{310\,\text{K}} \approx \frac{45{,}000}{8.314 \times 310^2} \approx 5.6\%\ \text{per °C} \tag{IX.3}$$

Propagation to F_k

Let $x = k_{\text{eff}}^{\text{surf}}[E]_t T$. The temperature sensitivity of F_k is:

$$\frac{\partial F_k}{\partial T_{\text{amb}}} = (1 - F_k)\,x\,\frac{E_a}{R\,T_{\text{amb}}^2} = (1 - F_k)\,|\ln(1 - F_k)|\,\frac{E_a}{R\,T_{\text{amb}}^2} \tag{IX.4}$$

At $F_k = 0.1$ (early signal, $x = 0.105$) and ±1°C uncertainty: $\delta F_k \approx 0.1 \times 0.105 \times 0.056 \approx 6 \times 10^{-4}$, a 0.6% absolute error on F_k. At $F_k = 0.5$ ($x = 0.693$): $\delta F_k \approx 0.5 \times 0.693 \times 0.056 \approx 1.9\%$.

Temperature control or measurement is required if absolute F_k accuracy $< 2\%$ is needed. The ratiometric signal $\Delta R/R_0$ is similarly affected since F_k enters linearly (Layer 13). The discrimination *ratio* F_k^c/F_k^h is less sensitive: at short times both F_k^c and F_k^h scale with the same $k_{\text{eff}}(T)$, so the ratio $[E]_c/[E]_h = 5\times$ is temperature-invariant.

X. Multi-Protease Competition

General competition model

Wound fluid contains multiple proteases that may cleave SGKIPRTLTA. In the first-order (linear) regime, contributions add independently:

$$\frac{dF_k}{dt} = k_{\text{obs,total}} = \sum_j k_{\text{eff},j}^{\text{surf}}\,[E_j]_t \tag{X.1}$$

The fractional contribution of interferent j to the observed cleavage signal is:

$$\xi_j = \frac{k_{\text{eff},j}^{\text{surf}}\,[E_j]_t}{k_{\text{obs,total}}} \tag{X.2}$$

MMP-9 selectivity requires $\xi_{\text{MMP-9}} \approx 1$, i.e. $k_{\text{eff,MMP-9}}^{\text{surf}}[E_{\text{MMP-9}}] \gg \sum_{j\neq\text{MMP-9}} k_{\text{eff},j}^{\text{surf}}[E_j]$.

Known interferents and status

Interferent	$[E_j]$ source	$k_{\text{eff},j}^{\text{surf}}$ status	Risk
HNE	30 nM chronic (Layer 0, Trengove 1999)	Unconfirmed	Moderate
MMP-2	Wound fluid present, conc. unconfirmed	Unconfirmed	Low–moderate
MMP-8	Wound fluid present, conc. unconfirmed	Unconfirmed	Low

Table 3: Known interferent proteases and their MMP-9 selectivity risk. None of the surface rate constants $k_{\text{eff},j}^{\text{surf}}$ are confirmed from papers in `papers/layer_09/`; HNE is the priority interferent.

No paper in `papers/layer_09/` *provides* k_{eff}^{surf} *for any interferent protease on SGKIPRTLTA. The competition fractions* ξ_j *cannot be computed until these are measured.* HNE is the highest-priority interferent given its confirmed elevated concentration in chronic VLU and its Ser-protease activity overlapping with some MMP substrates.

XI. Uncertainty Propagation

Analytic uncertainty in F_k

Let $x = k_{\text{eff}}^{\text{surf}}[E]_t T$, so $F_k = 1 - e^{-x}$. The partial derivatives are:

$$\frac{\partial F_k}{\partial k_{\text{eff}}} = \frac{x}{k_{\text{eff}}}\,e^{-x}, \qquad \frac{\partial F_k}{\partial [E]_t} = \frac{x}{[E]_t}\,e^{-x}, \qquad \frac{\partial F_k}{\partial T} = \frac{x}{T}\,e^{-x} \tag{XI.1}$$

By standard propagation:

$$\sigma_{F_k} = e^{-x}\,x\,\sqrt{\left(\frac{\sigma_{k_{\text{eff}}}}{k_{\text{eff}}}\right)^2 + \left(\frac{\sigma_{[E]}}{[E]_t}\right)^2 + \left(\frac{\sigma_T}{T}\right)^2} = (1-F_k)\,|\ln(1-F_k)|\,\sigma_{\text{rel,total}} \tag{XI.2}$$

Dominant uncertainty sources

Source	Relative uncertainty σ/μ	Status
$k_{\text{eff}}^{\text{surf}}$	Unknown; plausible range 2,000–67,500 $\text{M}^{-1}\text{s}^{-1}$ (factor 34×)	Dominant until measured
$[E]_t$	∼ 15–20% (Layer 0 ODE; healing estimate approximate)	Secondary
T (incubation)	∼ 1–2% (timer precision)	Negligible
T_{amb}	∼ 0.6% per °C (Eq. IX.3)	Tertiary

Table 4: Dominant uncertainty sources for the fractional cleavage F_k (Eq. (XI.2)). $k_{\text{eff}}^{\text{surf}}$ is the dominant source until the fabricated electrode is calibrated at Layer 15.

The dominant uncertainty is $k_{\text{eff}}^{\text{surf}}$, which spans more than an order of magnitude before electrode calibration. All F_k predictions are therefore presented as parametric sweeps over the plausible $k_{\text{eff}}^{\text{surf}}$ range. Once the fabricated electrode is measured (Layer 15 calibration), $k_{\text{eff}}^{\text{surf}}$ is pinned to ±5–10% and σ_{F_k} is determined by $\sigma_{[E]}$ alone.

XII. Parameter Table and Layer Outputs

Parameter	Des	Value	Source	Confirmed
$k_{\text{cat}}^{\text{sol}}$	Solu pha cat- alyt rate for SGI	86.4 s^{-1}	Kridel 2001 [2] Table II	Yes
K_m^{sol}	Solu pha Mic con- stan	1.28 mM	Kridel 2001 [2] Table II	Yes
$(k_{\text{cat}}/K_m)^{\text{sol}}$	Solu ef- fi- cien (UB for sur- face	67,500 $\text{M}^{-1}\text{s}^{-1}$	Kridel 2001 [2] Table II	Yes
$[E]_t^{\text{chronic}}$	Free MM 9 at 30- min reac chro	30 nM	Layer 0 ODE	Yes
$[E]_t^{\text{healing}}$	Free MM 9 at 30- min reac heal ing	6 nM	Layer 0 ODE	Yes
Γ_0	Init sur- face prol den- sity	$\sim$5 pmol cm^{-2}	Layer 7 (device-planned)	No
$k_{\text{eff}}^{\text{surf}}$	Surf rate con- stan (k_{ca} up- per bou	To be measured	Layer 15 calibration	No
$K_{m,\text{surf}}^{\Gamma}$	Surf MM con- stan (mo	6.4×10^{-13} mol cm^{-2} (estimate, Eq. IV.3)	Eq. IV.2; Layer 8 δ_{eff} unconfirmed	No
ρ	Pro satu ra- tio Γ_0/	0.3–3.1 (Table 1)	Layer 8 steric factor	No

Layer outputs.

- $F_k(T)$: fractional cleavage as function of incubation time → Layer 13 (ratiometric signal $\Delta R/R_0$)
- $k_{\text{obs}} = k_{\text{eff}}^{\text{surf}}[E]_t$: pseudo-first-order rate → Layer 14 (noise budget)
- T^*: optimal incubation time (Eq. (VIII.4)) → device protocol
- $k_{\text{eff}}^{\text{surf}}$ (post-calibration) → Layer 1 (dilution factor), Layer 15 (calibration curve)

XIII. Limitations

1. **$k_{\text{eff}}^{\text{surf}}$ is unknown; all F_k predictions are parametric.** The surface rate constant $k_{\text{eff}}^{\text{surf}}$ must be measured on the fabricated electrode before any absolute signal or LOD can be specified. The solution upper bound $k_{\text{cat}}^{\text{sol}}/K_m^{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ [2] is the starting assumption; surface immobilisation typically reduces this by 2–100× (Layer 8 steric correction, Adjémian 2010 [3]). Until calibration, the plausible range spans 2,000–67,500 $\text{M}^{-1}\text{s}^{-1}$ and T^* spans 16 min to > 9 h.

2. **K_M^{surf} and the LG zone assumption require experimental verification.** The $\lambda \approx 1$ conclusion holds for any $k_1 \leq 10^8\ \text{M}^{-1}\text{s}^{-1}$ and $[E]_t \leq 2.17$ nM. However, if the effective surface K_M^{surf} is pushed above ∼ 100 nM by steric crowding (putting $[E]_t/K_M^{\text{surf}}$ above 0.02), the system could enter a regime where the exact biexponential Eq. (III.7) is needed for accurate fitting. The diagnostic ratio $t_{1/2}/t_{1/4} \to 2.41$ (LG) vs. → 1.75 (FS) should be measured [1].

3. **Temperature dependence is unconfirmed.** The Arrhenius activation energy E_a for MMP-9 cleavage of SGKIPRTLTA on a surface is not reported in any paper in `papers/layer_09/`. The 5.6% per °C figure is a placeholder based on typical metalloproteinase values. A temperature-controlled incubation or in-situ temperature measurement is required for absolute F_k accuracy.

4. **Interferent cross-reactivity is unquantified.** The surface rate constants $k_{\text{eff},j}^{\text{surf}}$ for HNE, MMP-2, MMP-8, and other wound-fluid proteases on SGKIPRTLTA are not reported in any paper in `papers/layer_09/`. The selectivity fractions ξ_j cannot be computed. HNE is the priority interferent given its confirmed elevated chronic wound concentration (30 nM, Layer 0 [2]) and potential serine-protease activity on the substrate.

5. **Solution-to-surface K_m conversion requires δ_{eff} from Layer 8.** The effective surface Michaelis constant $K_{m,\text{surf}}^{\Gamma}$ is related to the solution value by $K_{m,\text{surf}}^{\Gamma} = K_m^{\text{sol}} \times 10^{-3} \times \delta_{\text{eff}}$, where δ_{eff} (Layer 8, unconfirmed) is the effective thickness of the accessible probe layer. Until δ_{eff} is established from Layer 8, $K_{m,\text{surf}}^{\Gamma}$ spans approximately 6×10^{-14}–6×10^{-12} mol cm^{-2} (δ_{eff} range 0.5–50 nm).

6. **The probe-saturation ratio ρ is unconfirmed; the progress curve shape is therefore uncertain.** Table 1 shows $\rho \in [0.3, 3.1]$ across the plausible steric correction range. Over this range, the progress curve transitions from near-monoexponential ($\rho \leq 0.3$) to significantly non-monoexponential ($\rho \geq 1$). The implicit integrated equation (Eq. (IV.5)) must be fit to measured progress curves to determine ρ empirically; fitting with the monoexponential alone will overestimate F_k at intermediate times if $\rho > 0.3$. This is the dominant source of *systematic* modelling error until Γ_0, δ_{eff}, and k_{cat} are measured on the fabricated electrode.

References

All papers in `electrochemical_device/papers/layer_09/`.

[1] Anne A, Demaille C (2012). Kinetics of enzyme action on surface-attached substrates: a practical guide to progress curve analysis in any kinetic situation. *Langmuir* 28:14665–14671. **anne_demaille2012.pdf** – ODE system (Eq. 1), exact biexponential solution (Eqs. 4–5), λ parameter (Eq. 6 region), universal progress curve (Eq. 7), FS limit (Eq. 8), zone diagram (Fig. 4): all confirmed from text.

[2] Kridel SJ, Chen E, Howard EW, Bhatt L, Strongin AY (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276:20572–20578. **kridel2001.pdf** – $k_{\text{cat}} = 86.4\ \text{s}^{-1}$, $K_m = 1.28$ mM, $k_{\text{cat}}/K_m = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ for SGKIPRTLTA (peptide A11), Table II confirmed.

[3] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing peptide-enzyme multilayer gold surfaces for the parallel, selective, and sensitive assay of trypsin and plasmin activities by electrochemical methods. *Langmuir* 26:10347–10356. **adjemian2010.pdf** – experimental surface cleavage yields and Γ_0 context.

[4] Squires TM, Messinger RJ, Manalis SR (2008). Making it stick: convection, reaction and diffusion in surface-based biosensors. *Nat Biotechnol* 26:417–426. **squires2008.pdf** – Damköhler number framework for reaction-limited vs. transport-limited biosensor regimes; surface MM ODE formulation.

[5] Pheeney CG, Barton JK (2012). DNA electrochemistry with tethered methylene blue. *Langmuir* 28:7063–7070. DOI: 10.1021/la300566x. **(in papers/layer_10/)** – MB surface coverage $\Gamma_0 = 1.3$–3.2 pmol cm^{-2} used for ρ calculation in Table 1.